AI, Rent-Seeking, and Ticket Allocation: Mechanism Design under Asymmetric Participation Costs

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A0 landscape · 1189 × 841 mm | Final Project working poster · verify official SDG artwork, full credits and final commit before release

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FINAL PROJECT A0 POSTER

KEY MESSAGE

AI lowers strategic participation costs asymmetrically. Synthetic tests show effort contests can cross a sharp threshold, while lotteries resist speed advantages but remain vulnerable to cheap identities.

01 Question and foundations

Research question & verified gap

RQ: How does AI cost asymmetry change genuine fan access and welfare?
Setting: 200 fans, 50 scalpers, one $100 ticket; resale $180.
Closest research: Leslie–Sorensen (2014), Hakimov et al. (2021), Mazorra–Della Penna (2023).
Gap: compare effort and verified randomization under an AI cost shock.

Strategic model and Intellectual Merits

Players: fans and scalpers; platform commits to a rule first.
Effort contest: $P_i = e_i / \Sigma e_j$; utility = $V_i P_i - c_i e_i$; active-set Nash benchmark.
Verified lottery: one fan entry; scalpers may buy up to 8 identities at cost k.
Contribution: compare robustness under asymmetric AI costs, not an empirical AI effect.

Three complementary lenses

Game theory — predicts effort under strategic cost changes.
Social choice — separates primary access, final ownership and welfare.
Mechanism design — contrasts effort-based allocation with verified randomization.

Auction or allocation comparison

Scarce resource: one representative concert ticket.
Contest: players pay effort costs regardless of outcome.
Lottery: random draw among entries; winner pays $100.
Both allow stipulated resale at $180. One ticket is NOT an entire concert inventory.

02 Model and verified evidence

INHERITED
Resale / rent-seeking
Booking and Sybil costs

OUR CONTRIBUTION
AI asymmetric cost shock
Contest vs. lottery

EVIDENCE
5,000-run synthetic tests
Sensitivity & welfare

Figure 1. Closest foundations [1–3] → bounded asymmetric-cost mechanism comparison → simulated outcomes. No real-market validation.

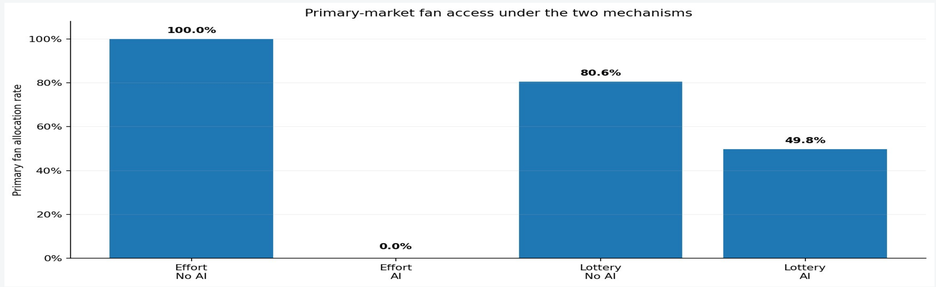


Figure 2. Synthetic primary fan allocation: Effort Contest 100.0% → 0.0%; Verified Lottery 80.6% → 49.8% (no AI → AI stress).
Seed 2026; 5,000 runs per main condition; 500 per sensitivity point. Reproduce: GitHub commit ce6cb2b.
Sensitivity: effort threshold at scalper costs 3→2; lottery identity cost k=.05 gives 7 identities and 36.2% fan access. Final ownership stays 100% in four baselines.

03 Practical impacts and roadmap

Practical Impacts: SDG 10 · Target 10.3

10 REDUCED INEQUALITIES

MECHANISM
Contest vs. lottery

INDICATOR
Primary fan access

Baseline (synthetic): 100.0% contest, 80.6% lottery.
Risk: privacy burden / false rejection; cheap identities.
Safeguards: appeals, auditability, accessible verification.
Proposed benefit ≠ demonstrated field impact.

Future Research Roadmap — NEW diagram

ECONOMICS | Estimate effort / identity costs
Method: platform data + sensitivity · Partner: applied economics
Criterion: stable estimates · Milestone: calibrated sweep

COMPUTER SCIENCE | Test adaptive scalpers
Method: rule-based ABM; optional LLM agents / OpenRouter
Partner: agent modelers · Criterion: reproducible logs · Milestone: ABM prototype

BEHAVIORAL SCIENCE | Test fairness perceptions
Method: structured pilot choices · Partner: behavioral researchers
Criterion: measured deviations · Milestone: pilot dataset

Integration, limitations and team contributions

Longfei Jing: strategic value and welfare accounting; symmetric hand check.
Tian Liang: exact Tullock computation and sensitivity; reproducibility checks.
Jiayang Sun: behavioral probe, poster integration, and primary-vs-final access interpretation.
All new ABM / LLM / pilot studies are PLANNED, not completed.

OCT 12 Literature & duet defense • OCT 14 Roundtable feedback • OCT 21 Final release

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Selected references

[1] Leslie & Sorensen (2014), Review of Economic Studies. doi:10.1093/restud/rdt033
[2] Hakimov et al. (2021), American Economic Review. doi:10.1257/aer.20191204
[3] Mazorra & Della Penna (2023), arXiv:2301.12813. doi:10.48550/arXiv.2301.12813
[4] Nash (1950), PNAS. doi:10.1073/pnas.36.1.48

OPEN MATERIALS

GitHub: github.com/dku-comsci-econ206-Autumn2026/PS2-Adaptive-Ticket-Auctions
Hugging Face: huggingface.co/spaces/dku-comsci-econ206-2026/ai-ticket-allocation
Baseline: ce6cb2b · Final commit to be recorded at release